# A river-network exposure model with a Whittle–Matérn breeding field for onchocerciasis prevalence in Sudan and South Sudan

## 1 Problem statement

Onchocerciasis is transmitted by blackflies of the genus *Simulium*, whose larvae develop in fast-flowing water. The exposure of a village is therefore generated at breeding sites on the river network, possibly several kilometres away, and reaches the village through the dispersal of flies, while the efficiency with which exposure is converted into infection depends on climate. In earlier work we expressed the force of infection as the product of an exposure computed from the river network and an intensity of transmission depending on aridity. With the suitability of a river described by its discharge alone, the fit required a quadratic effect of discharge and a change point in the effect of aridity, and both features were traced to a single focus beside the Nile rapids near Abu Hamad that discharge could not distinguish from slow stretches of the same river. Here we instead let the suitability of the rivers vary along the network through a Gaussian Whittle–Matérn field, add a village effect for residual variation, and simplify the fixed effects to a linear effect of log discharge on breeding and a linear effect of log aridity on the intensity of transmission.

## 2 Data

The outcome is the number of individuals with palpable onchocercal nodules among those examined in 900 villages of Sudan and South Sudan surveyed by the Rapid Epidemiological Mapping of Onchocerciasis (REMO). The river network is HydroRIVERS (Lehner and Grill, 2013), restricted to reaches with a long-term average discharge of at least 0.1 $\mathrm{m}^3/\mathrm{s}$; it contains 47,588 reaches and 249,403 km of river, and its topology, the reach into which each reach flows, is used to join the reaches at confluences. Aridity is the ratio of annual precipitation to reference evapotranspiration from version 3.1 of the Global Aridity Index (Zomer et al., 2022), averaged within 2 km of each village.

## 3 Model

Let $\Gamma$ denote the river network and $x_1, \ldots, x_m$ the villages, with $n_i$ individuals examined and $y_i$ found positive at $x_i$. The breeding intensity at a point $s \in \Gamma$ is

$$w(s) = \exp\{\theta \log Q(s) + u(s)\},$$

where $Q(s)$ is the discharge of the reach and $u$ is a zero-mean Gaussian field on $\Gamma$ describing the suitability of the river for breeding beyond what discharge explains. The exposure of a location $x$ is the breeding within reach, weighted by straight-line distance,

$$E(x) = \int_\Gamma \frac{1}{c\,\phi}\, k\!\left(\frac{\|x - s\|}{\phi}\right) w(s)\,\mathrm{d}s, \qquad k(t) = (1-t)_+^4(1+4t),$$

with the Wendland function $k$ (Wendland, 1995), $c = 2/3$ and the dispersal range fixed at $\phi = 40$ km from entomological evidence (Carnaghi et al., 2024). The intensity of transmission is

$$T(x) = \exp\{\beta_0 + \beta_1 \log \mathrm{AI}(x) + Z(x)\},$$

with independent village effects $Z(x_i) \sim N(0, \sigma_Z^2)$, and

$$Y_i \sim \mathrm{Bin}\big(n_i, 1 - \exp\{-E(x_i)\,T(x_i)\}\big).$$

The field $u$ is the Whittle–Matérn field on the metric graph $\Gamma$ with smoothness $\alpha = 1$ (Bolin et al., 2024), the solution of $(\kappa^2 - \Delta_\Gamma)^{1/2}\tau u = \mathcal{W}$ with Kirchhoff conditions at the vertices. Along a reach far from any vertex it is a stationary Ornstein–Uhlenbeck process with variance $\sigma_u^2$ and correlation $\exp(-\kappa d)$ at distance $d$ along the river, and we report its range as $\rho_u = 2/\kappa$, the distance at which this correlation is $e^{-2} \approx 0.14$. Because the field is Markov, its values at the vertices of the graph have a sparse precision matrix with an explicit form: an edge of length $\ell$ contributes

$$\frac{1}{\sigma_u^2(1 - e^{-2\kappa\ell})} \begin{pmatrix} (1 + e^{-2\kappa\ell})/2 & -e^{-\kappa\ell} \\ -e^{-\kappa\ell} & (1 + e^{-2\kappa\ell})/2 \end{pmatrix},$$

and subdividing an edge by additional vertices leaves the field unchanged. The Kirchhoff conditions make the variance twice $\sigma_u^2$ at the upstream ends of the network and smaller than $\sigma_u^2$ at confluences, so that information is shared across the reaches that meet at a confluence.

On the log scale, $\log \lambda(x) = \log E(x) + \log T(x)$. The two random terms enter different components and have different spatial structure: $u$ varies along the rivers and reaches a village only after smoothing by the dispersal kernel, so it is shared by the villages that draw on the same stretches of river, whereas $Z$ is specific to each village.

## 4 Estimation

The network is represented by a graph with a vertex at each end of every reach and additional vertices so that no edge is longer than 5 km, giving 75,119 vertices and 74,810 edges. The exposure integral is computed by quadrature over 148,571 nodes at most 2 km apart, and $u$ at each node is the linear interpolation of its values at the two ends of the edge. The latent vector, consisting of $\beta_0$, $\beta_1$, $\theta$, the values of $u$ at the vertices and the village effects, is integrated out by a Laplace approximation, with diffuse Gaussian priors (standard deviation 10) on the regression coefficients centred on their estimates without $u$ and $Z$. The mode is found by a truncated Newton method (Steihaug, 1983): the exact Hessian of the log posterior, which includes the curvature of $\log E$ in $\theta$ and $u$, is inverted by conjugate gradients preconditioned

by the Gauss–Newton Hessian. The latter is the prior precision plus a term whose rank is the number of villages, and is inverted by the Woodbury identity with one sparse Cholesky factor of the precision of $u$ for each value of the range; one evaluation of the approximate likelihood takes 10–20 seconds. The standard deviations $\sigma_u$ and $\sigma_Z$ are estimated by maximising the approximate marginal likelihood, and the range $\rho_u$ by profiling it over a grid. Standard errors of the regression coefficients are from the Gauss–Newton Hessian at the mode, conditional on the variance parameters. Computations use the R package `odispers`.

## 5 Results

Table 1 compares the models. Without latent effects, the linear model fits far worse than the earlier spline and quadratic specification (maximised log-likelihoods $-2851.53$ and $-2609.83$), which is why those features were needed. Adding the village effect raises the approximate log-likelihood to $-1613.40$, with a large standard deviation, $\sigma_Z = 2.21$, so most of the variation between villages is not explained by the fixed effects. Adding the breeding field raises it further, by 248 units at the best range, and halves the village effect to $\sigma_Z = 0.89$: much of the apparent heterogeneity between villages is variation along the rivers, shared by villages that draw on the same stretches. The profile over the range of $u$ (Figure 1) is maximal at about 800 km, with $\sigma_u = 2.76$. A range of several hundred kilometres along the network means that $u$ describes differences between rivers and sub-basins rather than between neighbouring reaches.

Table 1: Approximate (Laplace) log-likelihoods, with the regression coefficients integrated out under diffuse priors, and estimated standard deviations of the breeding field and of the village effect. Dispersal range $\phi = 40$ km in all models.

| Model | Log-likelihood | $\sigma_u$ | $\sigma_Z$ |
|---|---|---|---|
| Village effect only | $-1613.40$ | – | 2.21 |
| $u$, range 100 km, + village effect | $-1400.91$ | 2.80 | 0.80 |
| $u$, range 200 km, + village effect | $-1386.10$ | 2.65 | 0.81 |
| $u$, range 400 km, + village effect | $-1372.40$ | 2.55 | 0.86 |
| $u$, range 800 km, + village effect | $-1365.36$ | 2.76 | 0.89 |
| $u$, range 1600 km, + village effect | $-1368.70$ | 3.18 | 0.92 |

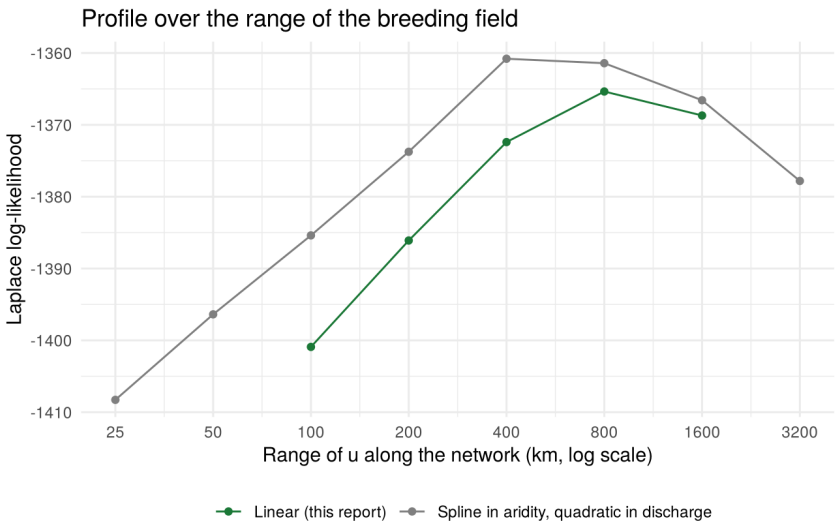



Figure 1: Approximate log-likelihood as a function of the range of the breeding field, for the linear model of this report and for the model with a spline in log aridity and a quadratic in log discharge.

Table 2 reports the regression coefficients. With the breeding field in the model, the effect of aridity is monotone and positive, $\beta_1 = 1.41$ (0.28): doubling the aridity index multiplies the force of infection by $2^{1.41} \approx 2.7$. The effect of discharge is negative, $\theta = -0.29$ (0.11), so that a tenfold increase in discharge roughly halves the breeding intensity per kilometre of river, consistent with small, fast-flowing streams being the main breeding habitat. The standard errors are two to five times larger than without latent effects, because they now account for the residual variation along the rivers and between villages.

Table 2: Estimates (standard errors) of the linear model. Maximum likelihood without latent effects; posterior mode and Gauss–Newton standard errors, conditional on the variance parameters, with the latent effects.

| | No latent effects | Village effect only | $u$ (800 km) + village effect |
|---|---|---|---|
| Intercept $\beta_0$ | −3.33 (0.20) | −3.34 (0.19) | −3.12 (0.37) |
| log AI, $\beta_1$ | 1.47 (0.05) | 1.20 (0.13) | 1.41 (0.28) |
| log $Q$, $\theta$ | −0.94 (0.13) | −0.23 (0.12) | −0.29 (0.11) |
| $\sigma_u$ | – | – | 2.76 |
| $\sigma_Z$ | – | 2.21 | 0.89 |

Figure 2 shows the mode of $u$ on the rivers within reach of a village. The largest values are on the Nile between Abu Hamad and the confluence with the Atbara, the stretch where the river crosses the Fourth and Fifth Cataracts and where the only infected desert villages are found, while the slow stretches of the Nile further north and south and the White Nile have negative values. In South Sudan the field is positive over the dense networks of small streams in the west and centre and in the far south-east. The field therefore identifies the river features that discharge cannot describe, and this is what allows the effects of discharge and aridity to be

linear: in the earlier model without $u$, the Abu Hamad focus produced the change point in the aridity effect, with a negative slope in the driest conditions, and the increase in breeding for the largest rivers.

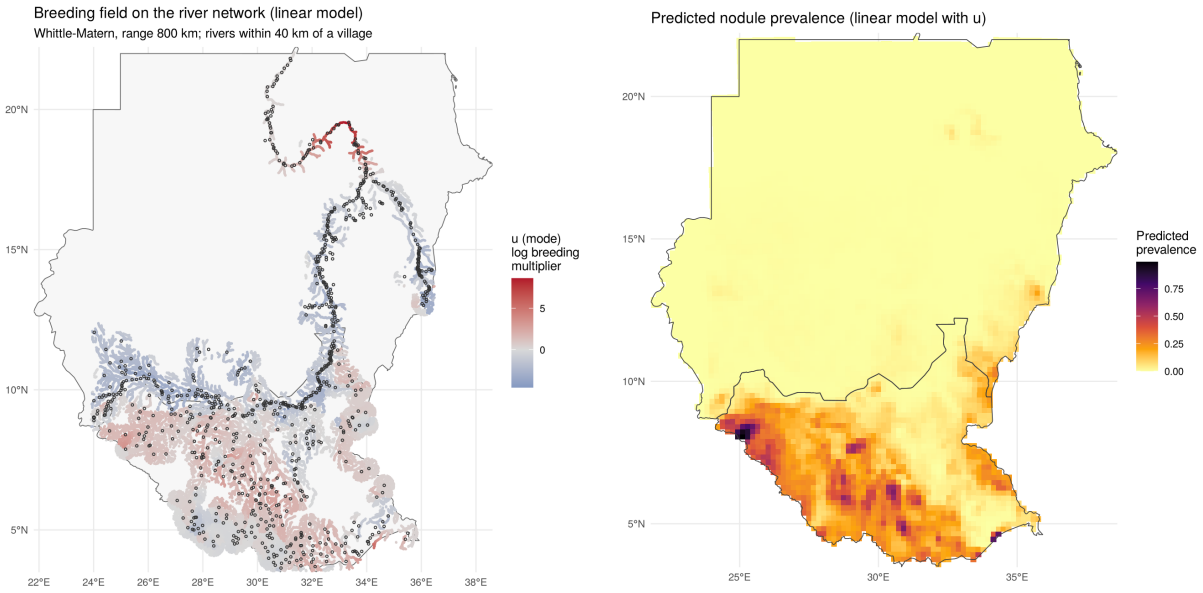



Figure 2: Left: mode of the breeding field $u$ (log multiplier of the breeding intensity) on the rivers within 40 km of a village; circles are the villages. Right: predicted prevalence on a 20 km grid, at the mode of $u$ and with the village effect set to zero.

The predicted prevalence (Figure 2, right) is close to zero across the desert and the Sahel, except along the Nile near Abu Hamad, and is highest in the south-west and centre of South Sudan and in the far south-east. Observed and predicted prevalence at the villages have a correlation of 0.88, against 0.58 for the linear model without latent effects (Figure 3), and the prevalence by aridity class is reproduced closely (Table 3). These are in-sample comparisons in which $u$ has been estimated from the same villages, so they overstate predictive performance; cross-validation is needed for an honest assessment. The predictions underestimate the highest prevalences, because they are evaluated at the mode with the village effect set to zero, which gives the median rather than the mean force of infection.

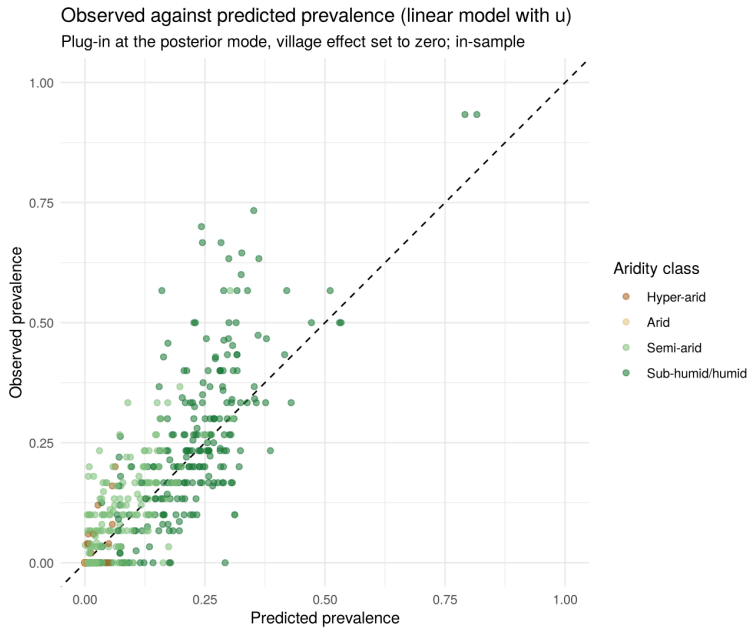



Figure 3: Observed against predicted prevalence at the 900 villages for the linear model with the breeding field (in-sample).

Table 3: Observed and predicted prevalence by aridity class (totals of positives over totals examined).

| Aridity class | Observed | Predicted, no latent effects | Predicted, with $u$ |
|---|---|---|---|
| Hyper-arid (AI $< 0.05$) | 0.005 | 0.001 | 0.003 |
| Arid (0.05–0.20) | 0.000 | 0.005 | 0.000 |
| Semi-arid (0.20–0.50) | 0.033 | 0.055 | 0.028 |
| Sub-humid and humid (AI $> 0.50$) | 0.252 | 0.197 | 0.228 |

If the breeding field accounts for the spatial structure in the data, the village effects $Z$ should be spatially independent once $u$ is in the model, and a further Gaussian process on the plane would not be needed. We assessed this with the empirical variogram of the predicted village effects, the posterior modes of $Z(x_i)$, in distance bins up to 300 km, and with an envelope of spatial independence obtained by randomly permuting the predicted effects among the village locations 999 times; the band between the 2.5th and 97.5th percentiles of the permuted variograms is the range expected in each bin when there is no spatial correlation. A global test combines the bins through the sum of squared standardised differences between the observed variogram and the mean of the permuted variograms, with the $p$-value given by the rank of the observed statistic among the permuted ones.

Figure 4 shows the result for the model with the village effect only and for the model with $u$. Without $u$, the variogram increases steadily from about 0.9 at short distances to 2.0 at 300 km and lies far below the envelope up to about 250 km, so that the village effects are strongly spatially correlated (global test $p = 0.001$, the smallest attainable with 999 permutations); in this model the village effect is absorbing the variation that $u$ represents. With $u$, the variance

of the predicted village effects falls from 2.17 to 0.18 and the variogram is flat and inside the envelope at every distance (global test $p = 0.155$). At the shortest distances it lies close to the upper limit of the envelope, which is the opposite of positive correlation: neighbouring villages draw on the same stretches of river and share $u$, so their own effects are estimated as deviations around it. We conclude that the residual variation between villages has no spatial structure, and keep $Z$ as an independent effect. Because the predicted effects are shrunk by different amounts depending on the number examined and on the information about $u$, they are not exactly exchangeable, so the permutation test is a diagnostic rather than a formal test of the model.

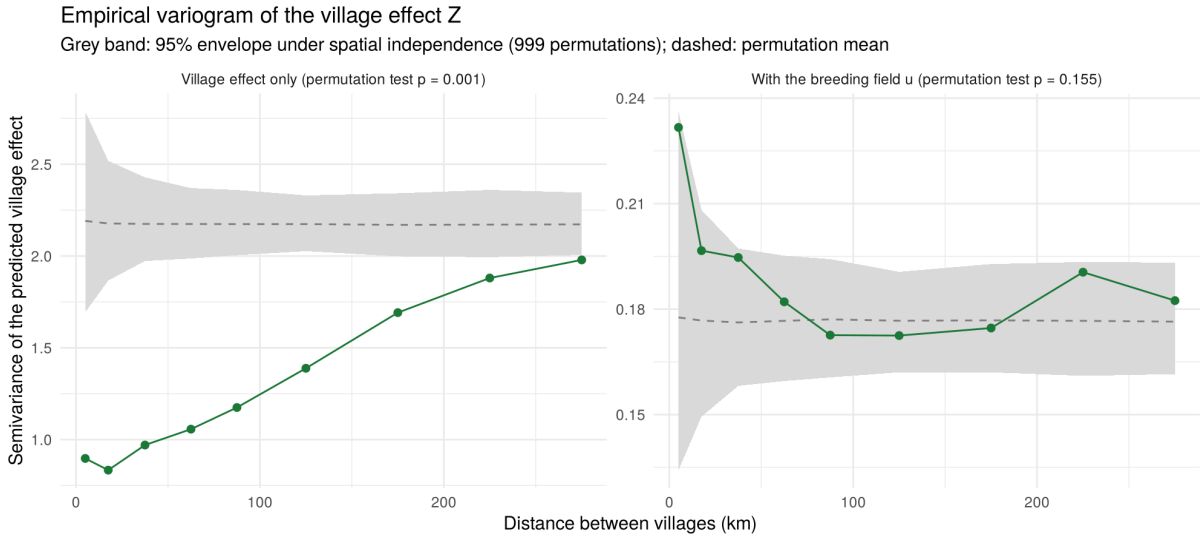



Figure 4: Empirical variogram of the predicted village effect $Z$ for the linear model with the village effect only (left) and with the breeding field $u$ (right). Grey band: 95% envelope under spatial independence from 999 random permutations of the predicted effects among the villages; dashed line: mean of the permuted variograms.

## 6 Zero inflation

We assessed the predictions of the model of the previous sections by ten-fold cross-validation with villages assigned to folds at random. In each fold the latent effects were refitted to the training villages, with the variance parameters and the range of $u$ fixed at their estimates from all the data, and for each held-out village the predictive distribution of the number of positives was obtained by combining the Gaussian approximation to the posterior of the latent effects, a new village effect and binomial sampling. The assessment was split in two parts, as in a hurdle model: the ability to separate villages without positives from villages with at least one, through the predicted probability $P(Y \geq 1)$, and the calibration of the predicted number of positives among the villages with at least one, through the non-randomised probability integral transform (PIT) of Czado et al. (2009) applied to the predictive distribution conditional on $Y \geq 1$. Under calibration the PIT histogram is flat; its expected variation was obtained by simulating counts from the predictive distributions.

The model separated zero and positive villages well (area under the ROC curve 0.94) but

predicted too few villages without positives, 480 against the 550 observed, almost all of the deficit being in the semi-arid class (219 expected against 275). Among villages with positives, the PIT histogram was hump-shaped, with too few values near 0 and 1 (Figure 6, left), so the predictive distributions were too wide. The two findings have a common explanation: the village effect, with a standard deviation of 0.89, must be wide enough to produce the many villages without positives, which makes it too wide for the villages where transmission occurs. This is the signature of zeros that arise from a separate process.

We therefore let each village lie inside or outside a zone of local transmission. Outside, no individual can be infected; inside, the number of positives is binomial as before. Both the probability $\pi(x)$ of lying in a zone and the force of infection $\lambda(x)$ within it are regressions on the logarithm of the exposure from the river network and on the logarithm of the aridity index, each with its own coefficients:

$$\operatorname{logit}\pi(x) = a_0 + a_E \log E(x) + a_1 \log \mathrm{AI}(x), \qquad \log\lambda(x) = b_0 + \gamma \log E(x) + b_1 \log \mathrm{AI}(x) + Z(x),$$

so that

$$P(Y_i = 0) = 1 - \pi(x_i) + \pi(x_i)\{1 - p(x_i)\}^{n_i}, \qquad P(Y_i = y) = \pi(x_i)\binom{n_i}{y} p(x_i)^y \{1 - p(x_i)\}^{n_i - y} \ (y > 0),$$

with $p(x) = 1 - \exp\{-\lambda(x)\}$. The exposure $E(x)$, with breeding linear in log discharge and the field $u$ (range 800 km), and the village effect $Z$ are as before. Rivers and climate may thus act both on whether transmission occurs and on how intense it is, and the data determine the balance. Where no river lies within reach, $E = 0$, so that $\pi = 0$ and $\lambda = 0$. For these fits the exposure is computed on the full river network of the two countries (478,526 km of river), so that predictions cover the whole area. The coefficients, the field and the village effects are integrated out by the same Laplace approximation, applied to the two predictors $\operatorname{logit}\pi(x_i)$ and $\log\lambda(x_i)$ of each village; for this likelihood the Hessian used in the approximation is the observed information of the two predictors, with eigenvalues bounded away from zero. The coefficients $a_E$ and $\gamma$ of $\log E$, which multiply a function of the latent field, are estimated together with $\sigma_u$ and $\sigma_Z$ by maximising the approximate marginal likelihood.

This formulation was reached in steps. A first zero-inflated model, $\pi(x) = 1 - \exp\{-\alpha E(x)\}$ with $\lambda(x) = E(x)^\gamma \exp\{b_0 + b_1 \log \mathrm{AI}(x) + Z(x)\}$, in which only the rivers determined the zones of transmission, already improved the approximate log-likelihood by 103 units and removed the deficiencies in calibration. The contrast between the arid north and the humid south, however, then had to be carried by the field $u$, and away from the surveyed rivers, where $u$ returns to its mean, the model predicted zones of transmission, and prevalences of several per cent, over arid parts of Darfur and Kordofan without survey data. Letting the climate enter the probability of a zone corrected this, and the regression form above makes the roles of rivers and climate in each part explicit. Table 4 compares the fits.

Table 4: Approximate (Laplace) log-likelihoods of the models with and without zero inflation, with the field $u$ (range 800 km) and the village effect. The last model is fitted on the full river network; the exposure at the villages is the same.

| Model | Log-likelihood | $\sigma_u$ | $\sigma_Z$ |
|---|---|---|---|
| No zero inflation | −1365.36 | 2.76 | 0.89 |
| Zero-inflated, rivers only in $\pi$: $\pi = 1 - e^{-\alpha E}$ | −1262.12 | 4.48 | 0.50 |
| Zero-inflated, rivers and aridity in $\pi$: $\pi = 1 - e^{-\alpha E \mathrm{AI}^{\delta}}$ | −1243.89 | 4.05 | 0.51 |
| Zero-inflated, $\log E$ and $\log \mathrm{AI}$ in both parts (logit, log) | −1237.12 | 7.51 | 0.47 |

Table 5 gives the estimates of the final model. The probability of lying in a zone of transmission increases steeply with humidity, $a_1 = 2.05$ (0.47), so that the odds of a zone are multiplied by $2^{2.05} \approx 4.1$ when the aridity index doubles, and with river exposure, $a_E = 0.58$, so that doubling the exposure multiplies the odds by $2^{0.58} \approx 1.5$. Within a zone, the force of infection increases more moderately with humidity, $b_1 = 0.58$ (0.11), and only weakly with exposure, $\gamma = 0.12$. Rivers and climate therefore act mainly on whether transmission occurs, and to a lesser degree on its intensity. The breeding intensity declines with discharge, $\theta = -0.59$ (0.33). The village effect is about half that of the model without zero inflation, $\sigma_Z = 0.47$ against 0.89, while the field $u$ is very variable, $\sigma_u = 7.5$, as it now maps the stretches of river that sustain transmission.

Table 5: Estimates (standard errors) of the zero-inflated model with $\log E$ and $\log \mathrm{AI}$ in both parts. Regression coefficients at the posterior mode, with standard errors conditional on $a_E$, $\gamma$ and the variance parameters, which maximise the approximate marginal likelihood. Approximate log-likelihood −1237.12.

| Part | Parameter | Estimate (s.e.) |
|---|---|---|
| Zone of transmission, $\mathrm{logit}\,\pi$ | intercept $a_0$ | 1.63 (0.67) |
| | $\log \mathrm{AI}$, $a_1$ | 2.05 (0.47) |
| | $\log E$, $a_E$ | 0.58 |
| Force of infection, $\log \lambda$ | intercept $b_0$ | −1.68 (0.14) |
| | $\log \mathrm{AI}$, $b_1$ | 0.58 (0.11) |
| | $\log E$, $\gamma$ | 0.12 |
| Breeding intensity | $\log Q$, $\theta$ | −0.59 (0.33) |
| Random effects | $\sigma_u$ (range 800 km) | 7.51 |
| | $\sigma_Z$ | 0.47 |

The estimated probability of lying in a zone of transmission (Figure 5) averages 0.06 for the hyper-arid villages, 0.01 for the arid, 0.33 for the semi-arid and 0.94 for the sub-humid and humid villages. It is close to zero along almost the whole of the Nile and the White Nile in Sudan and across the northern Sahel, high at the villages near Abu Hamad where infection was found in the desert, intermediate along the White Nile through the Sudd swamps, whose slow water offers little breeding habitat, and close to one over most of South Sudan. On the prediction grid

the expected prevalence $\pi(x)p(x)$ is essentially zero across the desert, also far from the surveys: of the 3,691 arid and hyper-arid grid cells, three have an expected prevalence above 5%.

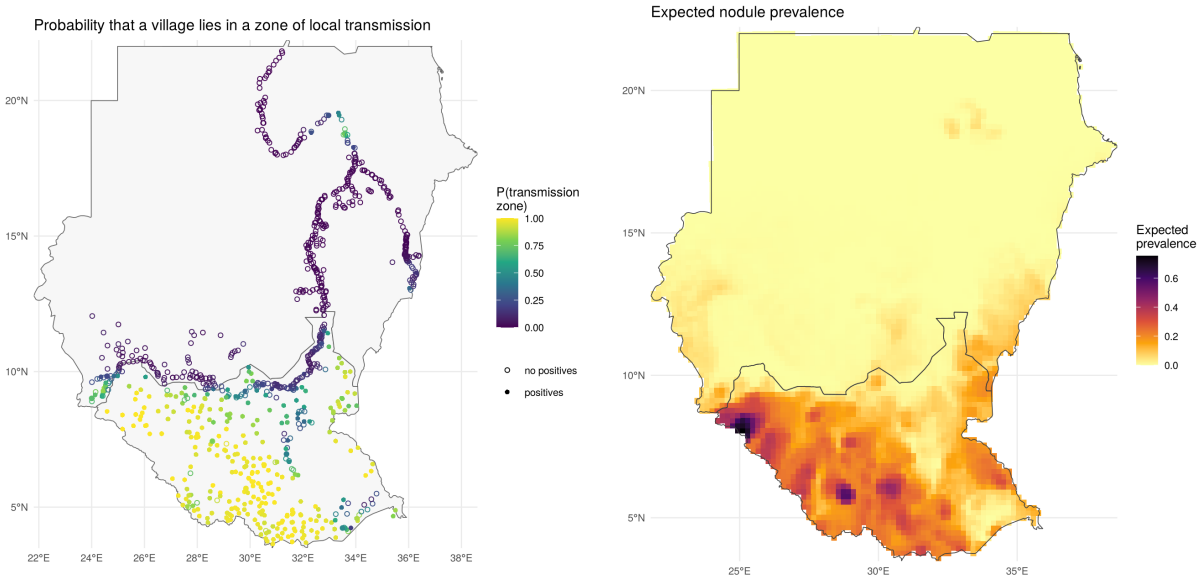



Figure 5: Zero-inflated model with $\log E$ and $\log \mathrm{AI}$ in both parts. Left: probability that each village lies in a zone of local transmission (filled symbols: villages with positives). Right: expected prevalence $\pi(x)p(x)$ on a 20 km grid, at the posterior mode with the village effect set to zero.

The random-fold cross-validation (Table 6 and Figure 6) shows that zero inflation removes both deficiencies of the model without it. Among villages with positives, the PIT histogram of the final model is close to flat, with nine of its ten bins within their expected variation, so the predicted numbers of positives are calibrated. The predicted number of villages without positives rises from 480 to 544, against 550 observed, and in the semi-arid class from 219 to 270, against 275; the reliability diagram of $P(Y \geq 1)$ lies close to the diagonal; and the area under the ROC curve, the Brier score and the mean logarithmic score all improve, the final model being the best of those considered on each of them. A modest excess of predicted zeros remains among the humid villages, 25 against 14 observed.

Table 6: Ten-fold cross-validation with random folds: discrimination between villages with and without positives, overall predictive score, and observed and expected numbers of villages without positives, in total and by aridity class.

| | No zero | Zero-inflated | | |
|---|---|---|---|---|
| | inflation | rivers in $\pi$ | rivers, aridity in $\pi$ | both parts |
| Area under the ROC curve | 0.939 | 0.943 | 0.947 | 0.949 |
| Brier score | 0.102 | 0.089 | 0.086 | 0.084 |
| Mean logarithmic score | −1.452 | −1.342 | −1.326 | −1.306 |
| Villages without positives (550) | 479.8 | 532.5 | 543.1 | 543.7 |
| Hyper-arid (154) | 144.0 | 141.1 | 148.5 | 149.1 |
| Arid (107) | 98.6 | 97.7 | 99.9 | 99.3 |
| Semi-arid (275) | 219.2 | 265.3 | 268.0 | 270.1 |
| Sub-humid and humid (14) | 18.1 | 28.5 | 26.7 | 25.2 |

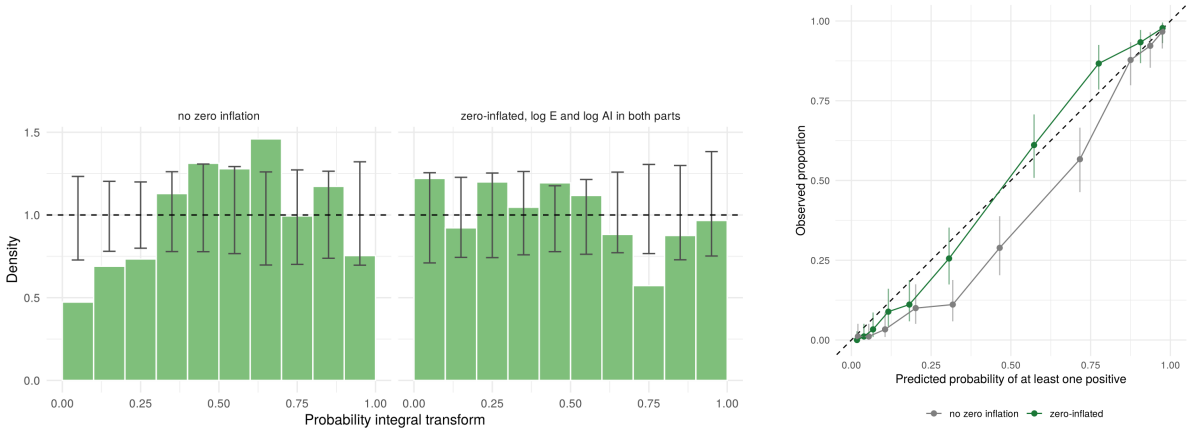



Figure 6: Ten-fold cross-validation with random folds, for the model without zero inflation and the zero-inflated model with $\log E$ and $\log \mathrm{AI}$ in both parts. Left: non-randomised PIT among villages with at least one positive; bars show the 95% range expected under calibration. Right: reliability of the predicted probability of at least one positive (deciles of the predicted probability, with 95% intervals).

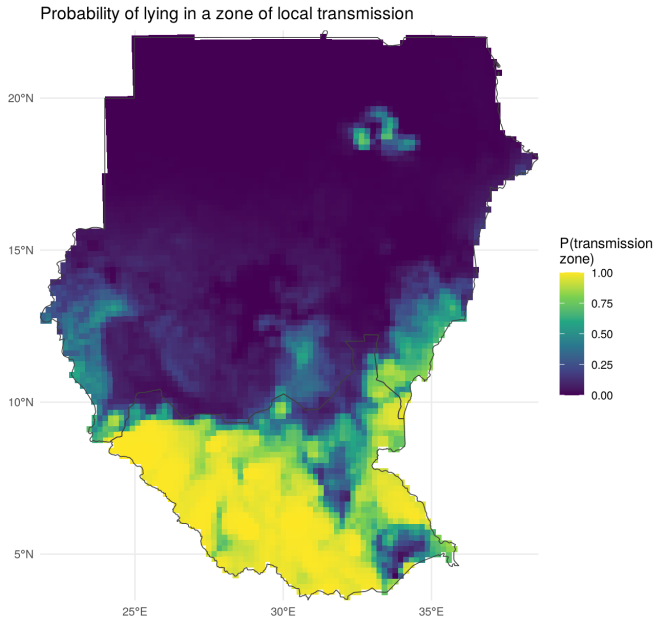



Figure 7: Probability of lying in a zone of local transmission on a 20 km grid (zero-inflated model with $\log E$ and $\log \mathrm{AI}$ in both parts, at the posterior mode).

# 7 Comparison with the more flexible specification and discussion

The same model with a spline in log aridity and a quadratic in log discharge reaches an approximate log-likelihood of $-1360.80$, at a range of 400 km, against $-1365.36$ for the linear model, a difference of 4.6 units for two additional coefficients and an estimated change point. Since the regression coefficients are integrated out, these marginal likelihoods include a penalty for each coefficient that depends on its prior standard deviation, so the comparison is indicative; but both specifications describe the data similarly once the breeding field is included. In the flexible specification with $u$, the slope of the aridity effect below the change point is $-0.8$ with a standard error of about 0.5 and the quadratic term in discharge is not significant, so its additional features are no longer supported. We therefore prefer the linear model, which has a monotone effect of aridity with a clear interpretation, a single parameter for the effect of river size, and attributes the remaining variation to the rivers and to the villages.

The decomposition of the force of infection into exposure and intensity of transmission is retained and sharpened. The exposure now depends on the rivers through both their discharge and the field $u$, which acts as a map of the suitability of the river network for breeding, and the intensity of transmission through aridity and a village effect. Places beyond 40 km of any river still have zero exposure. Several points need attention. First, $u$ is informed only on the rivers within reach of a village, so elsewhere its mode is close to zero and its uncertainty large; predictions in unsurveyed basins inherit this uncertainty, which the plug-in maps do not show. Second, the range of $u$ is large and its variance at the ends of the network is twice that

in the interior, as implied by the Kirchhoff conditions; a field with smoothness $\alpha = 2$ or a stationary boundary correction could be considered. Third, the field may be partly explained by measured river characteristics, such as channel gradient and the seasonality of flow from RiverATLAS, which would turn part of $u$ into interpretable covariates. Fourth, the absence of spatial correlation in the village effects once $u$ is included indicates that a Gaussian process on the plane in the intensity of transmission is not needed; such a process would in any case be difficult to separate from $u$, whose contribution to the exposure is smooth over several tens of kilometres after the dispersal kernel is applied. Fifth, the cross-validation shows that the excess of villages without positives is better described by a zero-inflated model, in which a village may lie outside any zone of transmission. With river exposure and aridity entering both the probability of a zone and the force of infection within it, the rivers and the climate act mainly on whether transmission occurs and less on its intensity, the predictions are calibrated, and transmission is not predicted in the desert away from the surveys. Finally, uncertainty in predictions should be obtained by sampling from the Laplace approximation rather than shown at the posterior mode.

## References


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